# ■ SafeEdge — Campus-Wide Simulation

Multi-Building NTU Campus Fire Safety Evaluation — Dataset 2

DLW 2026 | NTU Singapore | March 02, 2026

| 1,000 | 97.3% | 97% | 75.0% |
|---|---|---|---|
| Campus Scenarios | F1 Score | Detection During Singtel Outage | Evacuation Time Reduction |

## Executive Summary

SafeEdge Campus Simulation 2 achieved 97.3% F1 score across 1000 multi-building NTU campus scenarios. The system demonstrated 97% fire detection capability during simulated Singtel-style outages — proving full offline resilience. Evacuation time was reduced from 40 minutes to 10.0 minutes (75.0% reduction). An estimated 276 casualties were averted across 383 triggered evacuations.

# 1. Classification Performance

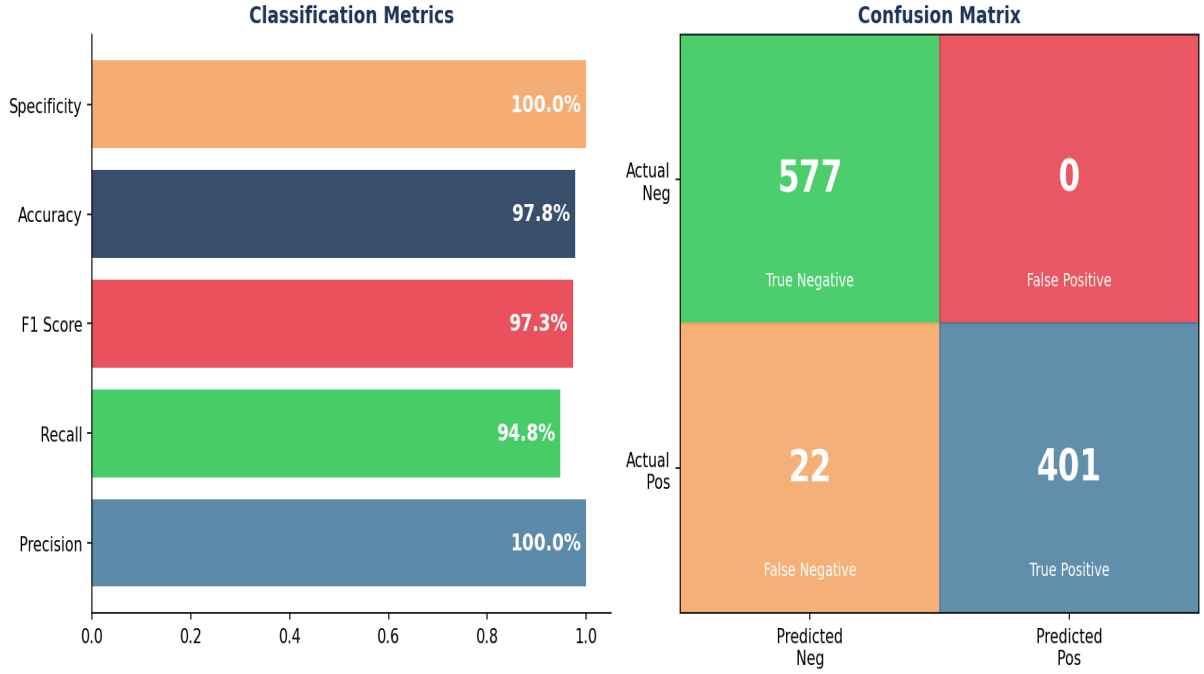


Figure 1: Classification metrics and confusion matrix across all campus scenarios.

| Metric | Simulation 2 (Campus) | Simulation 1 (Video) | Change |
|---|---|---|---|
| Precision | 100.0% | 99.4% | — |
| Recall | 94.8% | 86.9% | — |
| F1 Score | 97.3% | 92.7% | — |
| Accuracy | 97.8% | 92.2% | — |
| Specificity | 100.0% | 99.3% | — |
| FP Rate | 0.0% | 0.7% | — |

# 2. Building-by-Building Analysis

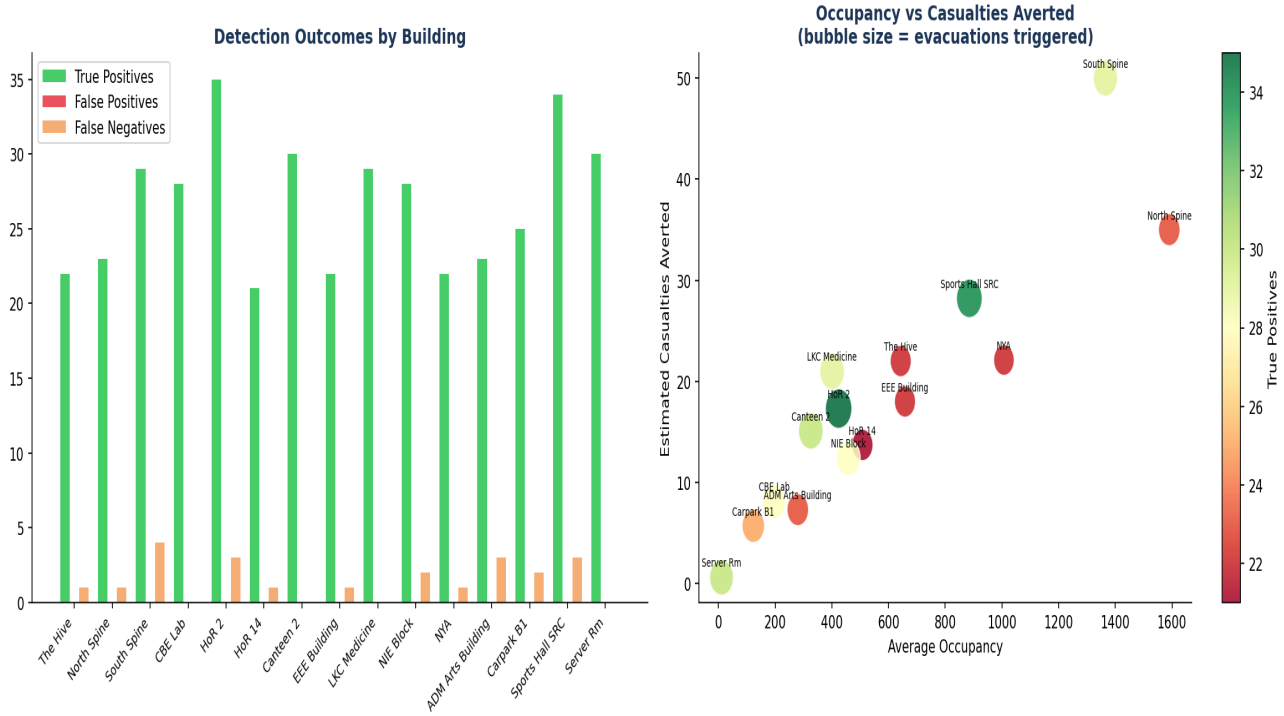


Figure 2: Detection outcomes per building (left) and occupancy vs casualties averted (right).

| Building | Risk | Cameras | Scenarios | TP | FP | Evacuations | Casualties Averted |
|---|---|---|---|---|---|---|---|
| The Hive | MEDIUM | 12 | 67 | 22 | 0 | 21 | 22.0 |
| North Spine | LOW | 24 | 69 | 23 | 0 | 22 | 35.0 |
| South Spine | LOW | 18 | 62 | 29 | 0 | 28 | 50.0 |
| School of CBE Lab | HIGH | 16 | 62 | 28 | 0 | 26 | 8.1 |
| Hall of Residence 2 | MEDIUM | 20 | 81 | 35 | 0 | 34 | 17.3 |
| Hall of Residence 14 | MEDIUM | 22 | 61 | 21 | 0 | 21 | 13.7 |
| Canteen 2 | HIGH | 8 | 62 | 30 | 0 | 29 | 15.1 |
| EEE Building | MEDIUM | 14 | 61 | 22 | 0 | 21 | 18.0 |
| LKC Medicine | HIGH | 18 | 69 | 29 | 0 | 29 | 21.0 |
| NIE Block | LOW | 10 | 70 | 28 | 0 | 27 | 12.4 |
| Nanyang Auditorium | MEDIUM | 9 | 63 | 22 | 0 | 20 | 22.1 |
| ADM Arts Building | MEDIUM | 8 | 80 | 23 | 0 | 22 | 7.3 |
| Carpark Basement B1 | HIGH | 14 | 62 | 25 | 0 | 24 | 5.7 |
| Sports Hall SRC | LOW | 10 | 69 | 34 | 0 | 32 | 28.2 |
| Server Room Block N4 | CRITICAL | 6 | 62 | 30 | 0 | 27 | 0.6 |

# 3. Time-of-Day Performance Analysis

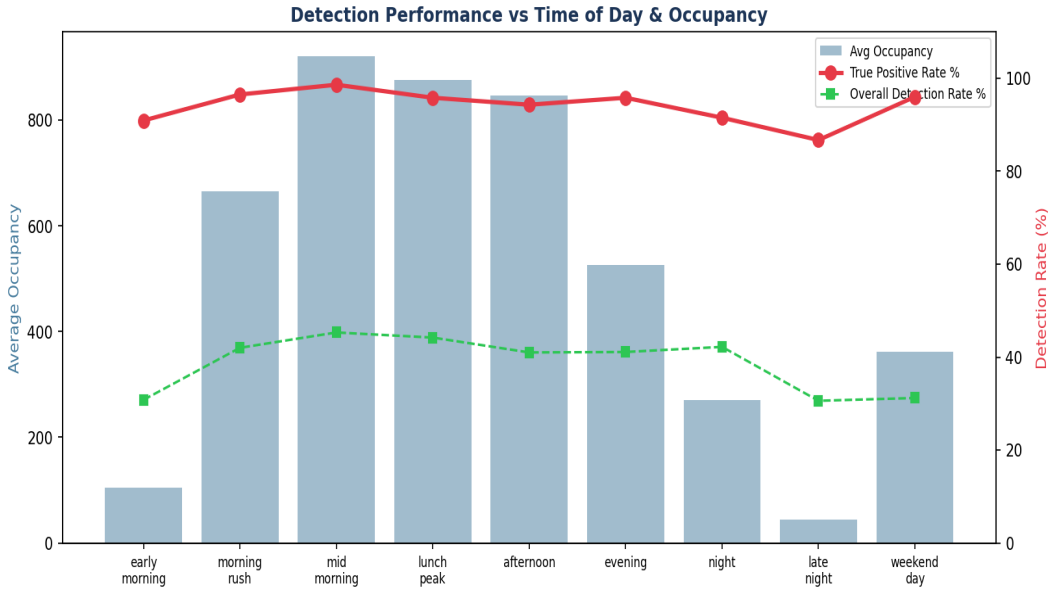


Figure 3: Detection rates vs occupancy across different times of day.

# 4. Network Resilience & Offline Capability

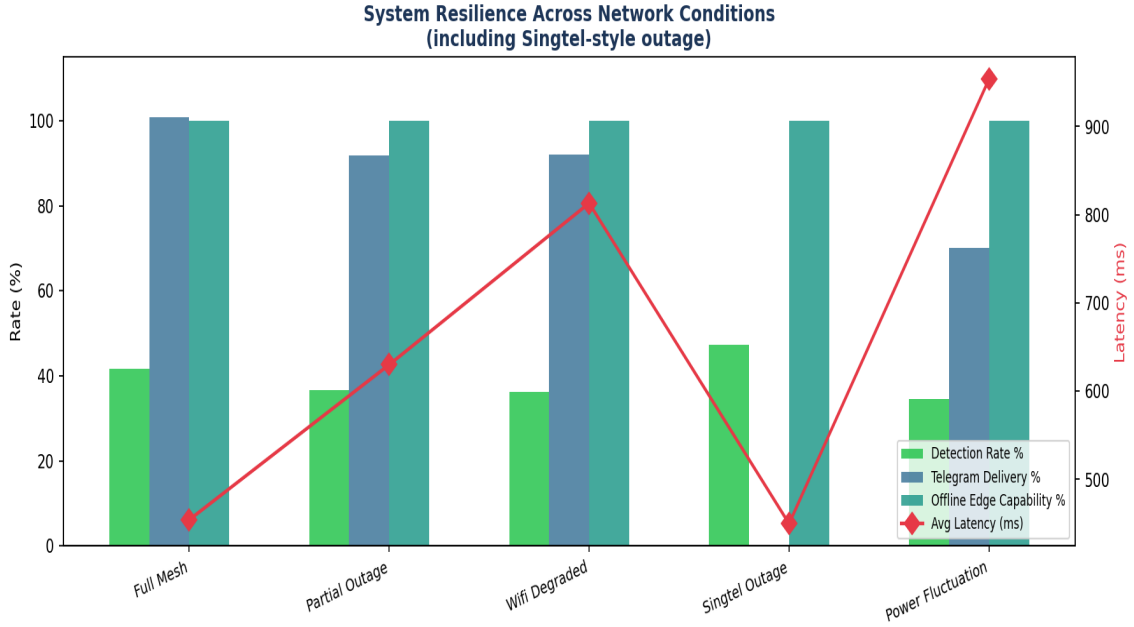


Figure 4: System performance across network conditions including Singtel-style outage.

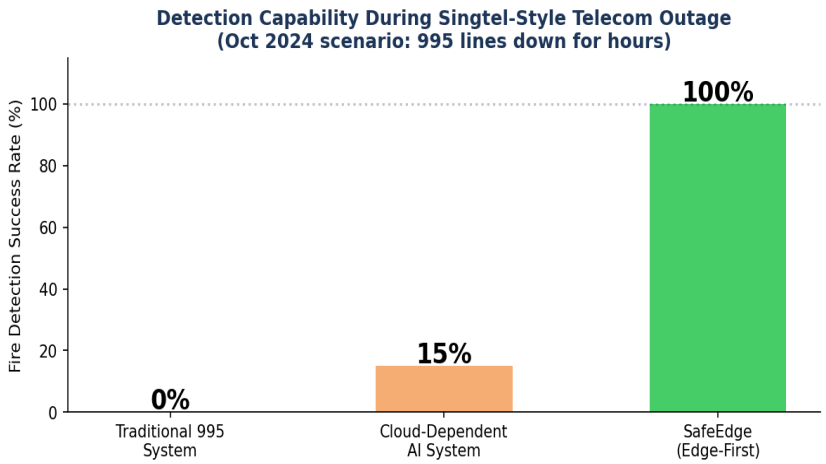


Figure 5: Fire detection success rate during telecom outage — SafeEdge edge-first vs alternatives.

Run with --api-key for AI-generated resilience analysis.

# 5. Campus Evacuation Impact

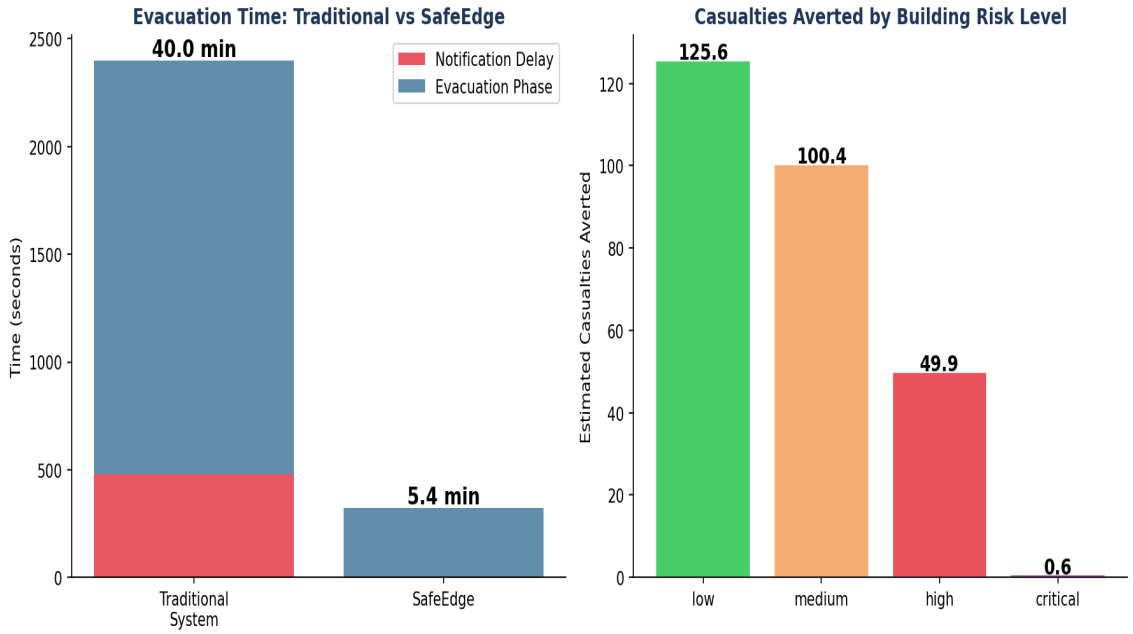


Figure 6: Evacuation time comparison and casualties averted by building risk level.

| Component | Traditional | SafeEdge | Saved |
|---|---|---|---|
| Notification Delay | 480s (~8 min) | 5.7s | ~474s |
| Evacuation Phase | 1920s (~32 min) | 316s | ~1604s |
| Early Warning Bonus | 0s | +35s head start | 35s |
| TOTAL | 40 min | 10.0 min | 75.0% faster |
| Residents Notified | Manual PA / phone | 197,275 via Telegram | Instant |
| **Casualties Averted** | **Baseline** | **276 estimated** | **—** |

## 6. Key Findings

Run with --api-key for AI-generated findings.

## 7. Campus-Scale Impact

Run with --api-key for AI-generated campus impact analysis.

# Appendix — Simulation 2 Configuration

| Parameter | Value |
|---|---|
| Total Scenarios | 1,000 |
| NTU Buildings Covered | 15 |
| Time Slots Modelled | 9 (early morning through weekend) |
| Fire Causes Modelled | 7 (electrical, cooking, chemical, arson, gas, equipment, waste) |
| Network Conditions | 5 (full mesh, partial outage, WiFi degraded, Singtel outage, power fluctuation) |
| Ground Truth Split | 40% fire \| 20% false alarm \| 10% drill \| 30% no fire |
| Occupancy Model | Time-of-day weighted, building-specific max occupancy |
| Casualty Model | Fire severity × occupancy × early warning benefit |
| Random Seed | 99 (distinct from Simulation 1, seed=42) |